A SOLUTION TO A PROBLEM OF ERDŐS-HERZOG-PIRANIAN

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Abstract. Given a monic polynomial f with zeroes in the interval $[-1,1]$ we calculate the infimum of the measure of the set $\{|f(x)|<1\}$, thus answering a question of Erdős, Herzog, and Piranian.

Contents

1. Introduction

A question of Erdős, Herzog, and Piranian from 1958 asks for the supremum and infimum of the measure of the set

$$E_f := \{x \in \mathbb{R} : |f(x)| < 1\}$$

when the zeros of a monic polynomial f are constrained to the interval $I = [-1,1]$ [6]. Though the original question asked more generally about the interval $[-r,r]$, later formulations by Erdős insist on the special case $I = [-1,1]$ [5], and this is how this question is often stated in the literature, including on the popular Erdős Problem list [2].

The supremum of $\mathcal{L}_f := |E_f|$ is $2\sqrt{2}$, as proved by [3, 4], in a series of technical papers. In [5] Erdős asked if a more elementary proof was possible, and this was recently supplied by Tao, who reformulated the problem in terms of logarithmic potentials of probability Borel measures μ on ([-1,1]) [11]. If

$$U_\mu(x) = \int_{\mathbb{R}} \log \frac{1}{|x-t|}\, d\mu(t), \qquad \mathcal{L}_\mu = \big|\{x \in \mathbb{R} : U_\mu(x) > 0\}\big|,$$

then the original polynomial problem corresponds to restricting μ to empirical measures μ_f associated with the zeros of monic polynomials f. This formulation enabled Tao to give a concise proof of the exact supremum in the measure theoretic setting:

$$\sup_{\mu} \mathcal{L}_\mu = 2\sqrt{2},$$

where μ runs over Borel propbability measures supported in $[-1, 1]$. He was also able to derive structural theorems for minimizers of the corresponding infimum problem. Based on these results, he conjectured that the optimal lower bound is approximately 1.835.

The present work confirms this conjecture of Tao, thus answering the question of Erdős, Herzog, and Piranian. Let D be the following constant, whose precise definition is given in Appendix A (see Definition A.2), as a solution to a system of non-linear equations:

$$D = 1.83443047576266171109075363512\,5\ldots$$

This constant D provides the optimal lower bound in case of both the measure theoretic and polynomial infimum problem:

Theorem 1.1. *If f runs over non-constant monic polynomials with all roots in $[-1, 1]$, and μ runs over Borel propbability measures supported in $[-1, 1]$ then*

$$\inf_f \mathcal{L}_f = \inf_\mu \mathcal{L}_\mu = D.$$

It was conjectured in [11] that $\inf_\mu \mathcal{L}_\mu$ is realized by a special Borel measure supported in $[-1, 1]$ and our work confirms this prediction as well.

Following the strategy suggested by [11], the difficult lower bound $L_f \geq D$ is proved using a duality argument. A hypothetical counterexample satisfying $L_f < D$ is put in a normal form with one distinguished positive component (ℓ, r) containing the atom -1 of μ_f. A short interval $[\alpha, \beta]$ is then selected to the right of that component. The positive part of U_{μ_f} inside this interval is denoted by T.

Then we design a dual measure λ with three properties:

(1) λ is supported at a point $z < \ell$, at the endpoint r, and on $[\alpha, \beta] \setminus T$;
(2) its potential U_λ is nonnegative on $[-1, 1]$;
(3) λ assigns positive mass to z, where $U_{\mu_f}(z) < 0$.

The first and third properties make $\int U_\mu \, d\lambda < 0$, while the second gives $\int U_\lambda \, d\mu \geq 0$. Since $\int U_\mu \, d\lambda = \int U_\lambda \, d\mu$, we arrive at a contradiction.

Related work. Recently this problem has seen remarkable progress, with many non-optimal lower bounds being proposed for $\inf_f \mathcal{L}_f$ and, in several cases, computer-certified. We record here the progress that we were able to trace through the comment section of the Erdős problem webpage [2].

On December 19, 2025, J. Koizumi claimed the bound 1.519 . . . This was followed by bounds of 1.614 and subsequently the golden ratio 1.618 . . . , proposed by J. Spier on December 31, 2025 and January 2, 2026, respectively. On January 2, Koizumi proposed the bounds 1.7 and later 1.75, while N. Sothanaphan reported numerical evidence for 1.7205. On January 3, Spier and Sothanaphan proposed the improved bounds 1.7877 and 1.7902, respectively, and on January 4 Spier announced the bound 1.8. More recently, H. Xu gave computer-assisted bounds 1.806 . . ., 1.807 . . ., and 1.814 . . . on May 2 and May 9, 2026, the last of these using a piecewise family of five-atom dual certificates. Finally, on June 10, 2026, Mendoza Lab announced the slightly stronger interval-arithmetic bound 1.814 . . ..

During the polishing of this manuscript, we learned that S. Wang had also independently generated a candidate proof of Erdős Problem 1308 using AI [14]. The approach of his argument is completely different from hours.

AI use. An initial version of the main argument was generated by GPT-5.5 Pro, using the prover–verifier framework described in [1]. This process began in late June, and the first complete AI-generated proof was obtained on July 11. Since then, we have verified, revised and simplified the argument, identifying appropriate attributions for arguments generated by the AI. Parts of this chronology can be verified from the timestamp of preliminary manuscripts uploaded to the Open Science Framework [12].

Symbolic and numeric calculation certificates. The proof uses three interval-arithmetic certificates and two exact symbolic verifiers. Their precise statements, software requirements, and execution order are recorded in Appendix C; the verifying Python scripts accompany the manuscript.

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2. Preliminaries

All measures in this work are assumed to be Borel measures. Beyond basic measure theoretic background, we only assume that the reader is familiar with the content of the short note by Tao [11]. We record well known results in this section, that will be needed later.

We first show that weakly converging measures induce L^1-converging potentials:

Lemma 2.1. *Let μ_j, μ be probability measures supported on $[-1,1]$, such that $\mu_j \rightharpoonup \mu$ weakly on $\mathbb{R}$. Then $U_{\mu_j} \to U_\mu$ in $L^1([-2,2])$.*

Proof. Choose $\chi \in C_c(\mathbb{R})$ such that $0 \le \chi \le 1$, $\chi = 1$ on $[-3,3]$, and put

$$g(t) := -\chi(t)\log|t|.$$

Since the logarithmic singularity is integrable, we have $g \in L^1(\mathbb{R})$.

For $M > 0$, define

$$g_M(t) := \chi(t)\min\{-\log|t|, M\}$$

Then g_M is bounded, continuous, and compactly supported, and

$$\|g_M - g\|_{L^1(\mathbb{R})} \to 0 \qquad \text{as } M \to \infty.$$

For a probability measure σ supported on $[-1,1]$, define

$$U_\sigma^{(M)}(x) := \int_{-1}^{1} g_M(x-y)\,d\sigma(y).$$

By weak convergence we obtain that $U_{\mu_j}^{(M)}(x) \to U_\mu^{(M)}(x)$ for every $x \in [-2,2]$. Moreover, $\left|U_{\mu_j}^{(M)}(x) - U_\mu^{(M)}(x)\right| \le 2\|g_M\|_\infty$. Therefore, by dominated convergence,

$$\left\|U_{\mu_j}^{(M)} - U_\mu^{(M)}\right\|_{L^1([-2,2])} \to 0, \quad \text{as} \quad j \to \infty.$$

We now remove the truncation. If $x \in [-2,2]$ and $y \in [-1,1]$ then $g(x-y) = \log\frac{1}{|x-y|}$. Thus, by Tonelli's theorem,

$$\left\|U_\sigma - U_\sigma^{(M)}\right\|_{L^1([-2,2])} \le \int_{-1}^{1}\int_{-2}^{2} |g(x-y) - g_M(x-y)|\,dx\,d\sigma(y) \le \|g - g_M\|_{L^1(\mathbb{R})}.$$

Consequently,

$$\begin{aligned}\left\|U_{\mu_j} - U_\mu\right\|_{L^1([-2,2])} &\le \left\|U_{\mu_j} - U_{\mu_j}^{(M)}\right\|_{L^1([-2,2])} + \left\|U_{\mu_j}^{(M)} - U_\mu^{(M)}\right\|_{L^1([-2,2])} + \left\|U_\mu^{(M)} - U_\mu\right\|_{L^1([-2,2])} \\ &\le 2\|g - g_M\|_{L^1(\mathbb{R})} + \left\|U_{\mu_j}^{(M)} - U_\mu^{(M)}\right\|_{L^1([-2,2])}.\end{aligned}$$

First letting $j \to \infty$ and then $M \to \infty$, we obtain $U_{\mu_j} \to U_\mu$, in $L^1([-2,2])$, as desired. ∎

Our next approximation result provides the strong link between the measure theoretic and polynomial version of the Erdős-Herzog-Piranian question.

Lemma 2.2. *For every probability measure μ supported on $[-1,1]$, there are empirical probability measures*

$$\mu_n = \frac{1}{N_n}\sum_{j=1}^{N_n}\delta_{x_{n,j}}, \qquad x_{n,j}\in[-1,1], \tag{2.1}$$

such that $\mathcal{L}_{\mu_n}\to\mathcal{L}_\mu$.

Proof. First we show that we can assume $|\{U_\mu=0\}|=0$. Reflection preserves $\mathcal{L}_\mu$ and sends discrete measures to discrete measures. We may therefore assume

$$\int_{-1}^{1} s\,d\mu(s)\le 0.$$

[11, Lemma 3.2] implies that

$$U_\mu(x)>0 \qquad (-1<x<0). \tag{2.2}$$

Outside $[-1,1]$, the potential is smooth and

$$U'_\mu(x)>0 \quad (x<-1), \qquad U'_\mu(x)<0 \quad (x>1).$$

It follows from these observations that, apart from a set of measure zero, every point of $\{U_\mu=0\}$ lies in $(0,1]$.

For $0<\varepsilon<1$, set

$$\mu_\varepsilon := (1-\varepsilon)\mu+\varepsilon\delta_{-1}. \tag{2.3}$$

Then

$$U_{\mu_\varepsilon}(x)=(1-\varepsilon)U_\mu(x)+\varepsilon\log\frac{1}{|x+1|}.$$

At almost every point where $U_\mu(x)=0$, we have $x\in(0,1]$, and therefore $\log\frac{1}{|x+1|}<0$. At almost every point where $U_\mu(x)\neq 0$, the sign of $U_{\mu_\varepsilon}(x)$ agrees with the sign of $U_\mu(x)$ for all sufficiently small ε. Consequently,

$$\mathbf{1}_{E_{\mu_\varepsilon}}(x)\to\mathbf{1}_{E_\mu}(x)$$

for almost every x as $\varepsilon\searrow 0$. Notice that all these positive sets lie in $(-2,2)$: if $|x|\ge 2$ and $y\in[-1,1]$, then $|x-y|\ge 1$, and therefore $U_\mu(x)\le 0, U_{\mu_\varepsilon}(x)\le 0$. So dominated convergence yields

$$\mathcal{L}_{\mu_\varepsilon}\to\mathcal{L}_\mu \qquad \varepsilon\searrow 0. \tag{2.4}$$

For almost every fixed $x\in(-2,2)$, the two quantities $U_\mu(x)$ and $\log(1/|x+1|)$ are finite. For such an x, the equation

$$(1-\varepsilon)U_\mu(x)+\varepsilon\log\frac{1}{|x+1|}=0$$

is an affine equation in ε, and hence has at most one solution unless both terms vanish. The latter can occur only where $\log\frac{1}{|x+1|}=0$, that is, at $x=-2$ or $x=0$. Fubini's theorem therefore implies that for almost every $\varepsilon\in(0,1)$,

$$\big|\{x\in\mathbb{R}:U_{\mu_\varepsilon}(x)=0\}\big|=0. \tag{2.5}$$

Choose $\varepsilon_k\searrow 0$ for which (2.5) holds.

Summarizing, we have $\mathcal{L}_{\mu_{\varepsilon_k}}\to\mathcal{L}_\mu$, and (2.5) holds for every k. Consequently, by a standard diagonal argument, it suffices to prove the result for measures satisfying

$$\big|\{U_\mu=0\}\big|=0.$$

We assume this condition for the remainder of the proof.

Now let μ_j be probability measures on $[-1,1]$, as defined in Equation (2.1). Basic measure theory implies existence of such a sequence satisfying $\mu_j \rightharpoonup \mu$ weakly.

From Lemma 2.1 we obtain that

$$U_{\mu_j} \to U_\mu \quad \text{in } L^1([-2,2]).$$

Fix $\varepsilon > 0$. If $x \in E_{\mu_j} \triangle E_\mu$ and $|U_\mu(x)| > \varepsilon$, then $U_{\mu_j}(x)$ and $U_\mu(x)$ have different signs, and therefore

$$|U_{\mu_j}(x) - U_\mu(x)| > \varepsilon.$$

It follows that

$$E_{\mu_j} \triangle E_\mu \subset \{|U_\mu| \le \varepsilon\} \cup \{|U_{\mu_j} - U_\mu| > \varepsilon\}.$$

Hence, by Markov's inequality,

$$\big||E_{\mu_j}| - |E_\mu|\big| \le |E_{\mu_j} \triangle E_\mu| \le |\{|U_\mu| \le \varepsilon\}| + \frac{1}{\varepsilon}\left\|U_{\mu_j} - U_\mu\right\|_{L^1([-2,2])}.$$

Taking limsup as $j \to \infty$ gives $\limsup_{j\to\infty} |E_{\mu_j} \triangle E_\mu| \le |\{|U_\mu| \le \varepsilon\}|$.

As $\varepsilon \searrow 0$, the sets $\{|U_\mu| \le \varepsilon\}$ decrease to $\{U_\mu = 0\}$. Since $|\{U_\mu = 0\}| = 0$, continuity from above of Lebesgue measure gives $|\{|U_\mu| \le \varepsilon\}| \to 0$. Therefore $|E_{\mu_j} \triangle E_\mu| \to 0$, as desired. □

The above result implies that the polynomial and measure theoretic notions of Theorem 1.1 agree:

Corollary 2.3. *If f runs over non-constant monic polynomials with all roots in $[-1,1]$ and μ runs over Borel propbability measures supported in $[-1,1]$ then*

$$\sup_f \mathcal{L}_f = \sup_\mu \mathcal{L}_\mu \qquad \text{and} \qquad \inf_f \mathcal{L}_f = \inf_\mu \mathcal{L}_\mu.$$

We shall use the following logarithmic-potential duality principle, that is based on [11, Lemma 2.1].

Lemma 2.4 (Logarithmic-potential duality)**.** *Let μ and λ be compactly supported probability measures. Assume that $U_\lambda(x) \ge 0$, for μ-almost every x, where the value $+\infty$ is allowed. Then*

$$\int_{\mathbb{R}} U_\mu \, d\lambda = \int_{\mathbb{R}} U_\lambda \, d\mu \ge 0. \tag{2.6}$$

In particular, $\int U_\mu \, d\lambda < 0$ is impossible.

Proof. Choose $R > 0$ such that $|x - t| \le R,\ x \in \operatorname{supp}\lambda,\ t \in \operatorname{supp}\mu$. Thus, $\log \frac{R}{|x-t|}$ is nonnegative on $\operatorname{supp}\lambda \times \operatorname{supp}\mu$, with the value $+\infty$ allowed on the diagonal. Tonelli's theorem therefore gives

$$\int_{\mathbb{R}}\int_{\mathbb{R}} \log\frac{R}{|x-t|}\, d\mu(t)\, d\lambda(x) = \int_{\mathbb{R}}\int_{\mathbb{R}} \log\frac{R}{|x-t|}\, d\lambda(x)\, d\mu(t).$$

Since both measures have total mass one, subtracting $\log R$ from the Tonelli identity yields (2.6).

The assumption $U_\lambda \ge 0$ μ-almost everywhere makes the right-hand side nonnegative, possibly $+\infty$, which proves the result. □

3. The geometry of a hypothetical counterexample

We begin by bringing a hypothetical counterexample to Theorem 1.1 into a certain normal form from [11]. To state this result, in addition to the optimal constant D, We shall also use a constant α, defined exactly in Appendix A, whose certified decimal expansion begins

$$\alpha = 0.804461754260998666\ldots.$$

Lemma 3.1 (Tao's normal form)**.** *Suppose that a finite atomic probability measure with support in $[-1,1]$ satisfies $\mathcal{L}_\mu < D$. After translation, reflection, and finitely many replacements of atoms by barycentres, one may assume the following.*

(i) *Every connected component of E_μ contains exactly one atom of μ.*
(ii) *One component is (ℓ, r), its atom is -1, and*

$$\operatorname{supp}\mu \subset \{-1\} \cup [r, 1], \qquad 0 \le r < \alpha, \qquad \ell \le -\sqrt{2}.$$

Proof. The barycentric-replacement argument of [11, Section 3] applies successively to the finite atomic measure μ and terminates after finitely many steps, without increasing $\mathcal{L}_\mu$. We may therefore assume that each component of E_μ contains exactly one atom. The remaining normalization and the bounds

$$\operatorname{supp}\mu \subset \{-1\} \cup [r, 1], \qquad r \ge 0, \qquad \ell \le -\sqrt{2}$$

are precisely those obtained in the discussion following [11, Lemma 3.2]. Lastly, if the estimate $r < \alpha$ failed to hold then $|E_\mu| \ge r - \ell \ge \alpha + \sqrt{2} \approx 2.218 > D$, a contradiction. □

The next lemma is the first numerically certified fact in the argument. The result and argument is due to user 'jspier', who described it in the comment section of [2] on January 4, 2026.

Lemma 3.2. *In the normal form of Lemma 3.1, one has*

$$\ell \le -1.708.$$

Proof. Assume for contradiction that $\ell > -1.708$. By Lemma 3.1,

$$-1.708 < \ell \le -\sqrt{2}.$$

For every

$$b \in [0, \le 1.836 + \ell]$$

define the finite measure

$$\nu_{\ell,b} := \delta_\ell + (1.395 - b)\delta_b + \Gamma(\ell, b)\delta_{1.071-b},$$

where

$$\Gamma(\ell, b) := \frac{-10^{-4} - \log(-1-\ell) - (1.395 - b)\log(1 + b)}{\log(2.071 - b)}. \tag{3.1}$$

This choice of $\Gamma(\ell, b)$ gives the exact identity

$$U_{\nu_{\ell,b}}(-1) = \log\frac{1}{-1-\ell} + (1.395 - b)\log\frac{1}{1+b} + \Gamma(\ell, b)\log\frac{1}{2.071 - b} = 10^{-4}. \tag{3.2}$$

The script `erdos1038_forcing_exact.py`, recorded in Appendix C, uses outward-rounded Arb arithmetic to certify that for

$$-1.708 \le \ell \le -\sqrt{2}, \qquad 0 \le b \le 1.836 + \ell, \qquad 0 \le x \le 1,$$

the following numerical estimates hold:

$$\Gamma(\ell, b) > 0, \qquad U_{\nu_{\ell,b}}(x) > 0. \tag{3.3}$$

It also certifies the elementary inequalities needed in the measure-counting step:

$$1.836 + \ell > 0, \qquad 1.395 - b > 0, \qquad 2(1.836 + \ell) < 1.071. \tag{3.4}$$

Let

$$M_{\ell,b} := \nu_{\ell,b}(\mathbb{R}) = 1 + (1.395 - b) + \Gamma(\ell, b) > 0, \qquad \widehat{\nu}_{\ell,b} := M_{\ell,b}^{-1}\nu_{\ell,b}.$$

Thus $\widehat{\nu}_{\ell,b}$ is a probability measure. The normal form gives

$$\operatorname{supp}\mu \subset \{-1\} \cup [r, 1], \qquad r \ge 0.$$

Hence (3.2) and (3.3) imply that $U_{\widehat{\nu}_{\ell,b}} > 0$ at every atom of μ, with the value $+\infty$ allowed if an atom of μ coincides with an atom of $\widehat{\nu}_{\ell,b}$. Applying Lemma 2.4 with $\lambda = \widehat{\nu}_{\ell,b}$, we obtain

$$0 < \int_{\mathbb{R}} U_{\widehat{\nu}_{\ell,b}}\, d\mu = \int_{\mathbb{R}} U_\mu\, d\widehat{\nu}_{\ell,b}.$$

Multiplying this by $M_{\ell,b}$ gives

$$0 < U_\mu(\ell) + (1.395 - b)U_\mu(b) + \Gamma(\ell, b)U_\mu(1.071 - b). \tag{3.5}$$

The point ℓ is a non-atomic endpoint of the component (ℓ, r), so continuity of U_μ gives $U_\mu(\ell) = 0$. Since the two remaining coefficients in (3.5) are positive, we conclude that for every $0 \leq b \leq 1.836+\ell$,

$$b \in E_\mu \qquad \text{or} \qquad 1.071 - b \in E_\mu. \tag{3.6}$$

Set

$$I := [0, 1.836 + \ell], \qquad J := [1.071 - (1.836 + \ell), 1.071],$$

and let $A := E_\mu \cap I$. By (3.4), the intervals I and J are disjoint. For every $b \in I \setminus A$, (3.6) gives $1.071 - b \in E_\mu \cap J$. Since the reflection $b \mapsto 1.071 - b$ preserves Lebesgue measure,

$$|E_\mu \cap J| \geq |I \setminus A|.$$

Consequently,

$$|E_\mu \cap (I \cup J)| = |A| + |E_\mu \cap J| \geq |A| + |I \setminus A| = 1.836 + \ell.$$

Finally, $(\ell, 0) \subset (\ell, r) \subset E_\nu$, and $(\ell, 0)$ is disjoint from $I \cup J$. Therefore

$$\mathcal{L}_\mu \geq |(\ell, 0)| + |E_\mu \cap (I \cup J)| \geq -\ell + (1.836 + \ell) = 1.836.$$

The certified value for D obtained in Appendix A gives $D < 1.836$, contradicting $\mathcal{L}_\mu < D$. Hence $\ell \leq -1.708$. ∎

Lemma 3.3. *With the notation and terminology of Lemma 3.1 there exists $h \geq 0$ such that for*

$$\beta := 1 + \frac{h}{2} \quad \text{and} \quad z := r + h - D,$$

the following properties hold:

$$h = |E_\mu \cap [\alpha, \beta]|, \qquad \beta \notin E_\mu, \tag{3.7}$$

$$r + h < D - 1.708, \qquad z < \ell. \tag{3.8}$$

Moreover,

$$T := \operatorname{int}\big(E_\mu \cap [\alpha, \beta]\big)$$

is a finite union of disjoint open intervals of total length h.

The relation $\beta = 1 + h/2$ balances the length h of the exceptional set against the amount by which the window $[\alpha, \beta]$ extends beyond 1. The point $z = r + h - D$ is then a point that falls to the left of ℓ if the total positive length is less than D.

Proof. Since the component (ℓ, r) is disjoint from $[\alpha, \infty)$,

$$(r - \ell) + |E_\mu \cap [\alpha, \infty)| \leq \mathcal{L}_\mu < D.$$

Using $\ell \leq -1.708$ from Lemma 3.2, we obtain

$$|E_\mu \cap [\alpha, \infty)| < D - r - 1.708.$$

The function

$$b \longmapsto b - 1 - \frac{1}{2}|E_\mu \cap [\alpha, b]|, \qquad b \geq 1,$$

is continuous (in fact Lipschitz). At $b = 1$ it is nonpositive, while at

$$b = 1 + \frac{1}{2}|E_\mu \cap [\alpha, \infty)|$$

it is nonnegative. Hence it vanishes at some β between these two points. Put $h = |E_\mu \cap [\alpha, \beta]|$. Then $\beta = 1 + h/2$, and

$$h \leq |E_\mu \cap [\alpha, \infty)| < D - r - 1.708, \tag{3.9}$$

which proves the first inequality in (3.8).

We claim that $\beta \notin E_\mu$. If $\beta = 1 \in E_\mu$, openness would give positive length in $E_\mu \cap [\alpha, 1]$, contradicting $h = 0$. Thus a counterexample to the claim would have $\beta > 1$. Since $\operatorname{supp} \mu \subset [-1, 1]$,

$$U_\mu'(x) = -\int \frac{d\mu(t)}{x - t} < 0 \qquad (x > 1).$$

Therefore $U_\mu(\beta) > 0$ would imply $(1, \beta] \subset E_\mu$. For $x \geq 1$ and $t \leq 1$,

$$|2 - x - t| \leq x - t,$$

so $U_\mu(2 - x) \geq U_\mu(x)$. It follows that $[2 - \beta, 1) \subset E_\mu$, including the point $2 - \beta$. By (3.9),

$$2 - \beta = 1 - \frac{h}{2} > 1 - \frac{D - 1.708}{2} \approx 0.937 > \alpha.$$

Thus $(2 - \beta, \beta) \subset E_\mu \cap [\alpha, \beta]$, and this interval already has length $2(\beta - 1) = h$. Openness at $2 - \beta$ supplies additional positive length, a contradiction. Hence $\beta \notin E_\mu$.

Finally,

$$D > \mathcal{L}_\mu \geq (r - \ell) + h,$$

so $z = r + h - D < \ell$. The statement about T follows because E_μ is a finite union of open intervals and endpoints have measure zero. □

4. The dual measure λ and the shape of its potential

Motivated by the findings of the previous section, for this section we fix numbers

$$r \geq 0, \qquad h \geq 0, \qquad r + h \leq D - 1.708, \tag{4.1}$$

and put

$$\beta := 1 + \frac{h}{2}, \qquad z := r + h - D.$$

Then $z \leq -1.708 < -1$, $r < \alpha < \beta$, and $h < 2(1 - \alpha)$. Let

$$T = \bigcup_{j=1}^{m} (u_j, v_j) \subset [\alpha, \beta]$$

be a finite union of disjoint open intervals, ordered from left to right; the empty union is allowed. Recall also that

$$D = 1.834430475762661711090\ldots, \qquad \alpha = 0.804461754260998666\ldots.$$

For $a < b$, let

$$\chi_{a,b}(w) := \exp\left(\frac{1}{2} \operatorname{Log} \frac{w - a}{w - b}\right) = \sqrt{\frac{w - a}{w - b}}, \qquad w \in \mathbb{C} \setminus [a, b],$$

where Log is the principal logarithm. This can be done, since $\frac{w-a}{w-b} \notin (-\infty, 0]$ for $w \in \mathbb{C} \setminus [a, b]$. Thus $\chi_{a,b}(w) \to 1$ as $w \to \infty$.

Now we introduce

$$C \in [1, -z),$$

and define

$$S_{T,C}(w) = \frac{w + C}{(w - z)(w - r)} \chi_{\alpha,\beta}(w) \prod_{j=1}^{m} \chi_{u_j,v_j}(w). \tag{4.2}$$

Definition 4.1 (The dual measure)**.** On $(\alpha,\beta)\setminus T$, we give $\lambda_{T,C}$ the density

$$\rho_{T,C}(s)ds := \frac{1}{\pi}\frac{s+C}{(s-z)(s-r)}\sqrt{\frac{s-\alpha}{\beta-s}}\prod_{j=1}^{m}\sqrt{\left|\frac{s-u_j}{s-v_j}\right|}\,ds. \tag{4.3}$$

We add atoms at z and r whose masses are the residues of $S_{T,C}$ there ($\operatorname{Res}_z S_{T,C}$ and $\operatorname{Res}_r S_{T,C}$). If the last component of T ends at β, we also add the atom at β whose mass is $\operatorname{Res}_\beta S_{T,C}$.

The density of Equation (4.3) and all the weights of the point masses are non-negative. At z and r, every square-root factor is positive, and the signs of $z+C$, $z-r$, $r+C$, and $r-z$ give positivity of $\operatorname{Res}_z S_{T,C}, \operatorname{Res}_r S_{T,C}$. If the component (u_m,β) is present in T, the endpoint residue is

$$\frac{\beta+C}{(\beta-z)(\beta-r)}\sqrt{(\beta-\alpha)(\beta-u_m)}\prod_{j<m}\sqrt{\frac{\beta-u_j}{\beta-v_j}} > 0.$$

The dual measure $\lambda_{T,C}$ is most naturally understood through its Cauchy transform:

Proposition 4.2 (Cauchy transform)**.** *The measure $\lambda_{T,C}$ is a positive probability measure, and*

$$\int_{\mathbb{R}}\frac{d\lambda_{T,C}(s)}{w-s} = S_{T,C}(w) \tag{4.4}$$

for $w\in\mathbb{C}\setminus\operatorname{supp}\lambda_{T,C}$. Hence, on real intervals disjoint from $\operatorname{supp}\lambda_{T,C}$,

$$U'_{\lambda_{T,C}}(x) = -S_{T,C}(x). \tag{4.5}$$

Additionally, the potential $U_{\lambda_{T,C}}$ is constant on every component of $(\alpha,\beta)\setminus T$.

Proof. For $s\in(\alpha,\beta)\setminus T$, the upper and lower boundary values satisfy

$$S_{T,C,+}(s) - S_{T,C,-}(s) = -2\pi i\,\rho_{T,C}(s), \tag{4.6}$$

where $\rho_{T,C}$ is the density in (4.3). Indeed, for $u<v$ and $s\in(u,v)$,

$$\chi_{u,v,+}(s) = -i\sqrt{\frac{s-u}{v-s}}, \qquad \chi_{u,v,-}(s) = +i\sqrt{\frac{s-u}{v-s}}.$$

On an interval $(u_j,v_j)\subset T$, the jump contributions from χ_{u_j,v_j} and $\chi_{\alpha,\beta}$ cancel, so $S_{T,C,+}-S_{T,C,-}=0$ on T.

We now prove (4.4). Fix $w\notin\{z,r\}\cup[\alpha,\beta]$, and consider

$$F_w(\zeta) = \frac{S_{T,C}(\zeta)}{\zeta-w}.$$

The function $S_{T,C}$ is holomorphic off the cut $[\alpha,\beta]$, except for the simple poles at z and r, and possibly a simple pole at β when the last component of T has right endpoint β. Take a large circle, cut the plane along $[\alpha,\beta]$, and remove small circles around all branch points, around the simple poles, and around w. Apply the residue theorem on this slit domain to F_w, then let the small circle radii tend to zero and the large radius tend to infinity.

The large-circle integral tends to zero because

$$S_{T,C}(\zeta) = \frac{1}{\zeta} + O(\zeta^{-2}), \qquad F_w(\zeta) = O(\zeta^{-2}).$$

Near a branch endpoint e, the transform has at worst a square-root singularity, $S_{T,C}(\zeta) = O(|\zeta - e|^{-1/2})$, so the integral around a circle of radius ε is $O(\varepsilon^{1/2})$ and tends to zero. The remaining contributions come from the pole at w, the poles corresponding to the atoms, and the jump across the cut $[\alpha,\beta]$. Moving the residue at w to the other side gives

$$S_{T,C}(w) = \sum_p \frac{\operatorname{Res}_p S_{T,C}}{w-p} + \frac{1}{2\pi i}\int_\alpha^\beta \frac{S_{T,C,+}(s)-S_{T,C,-}(s)}{s-w}\,ds,$$

where $p = z, r$, and also $p = \beta$ when the endpoint atom is present. The jump formula (4.6) (and the comments under it) turns the last term into

$$\int_{(\alpha,\beta)\setminus T} \frac{\rho_{T,C}(s)}{w-s}\, ds.$$

This gives exactly (4.4) for $w \notin \{z, r\} \cup [\alpha, \beta]$. Extension to every $w \notin \operatorname{supp} \lambda_{T,C}$ follows by holomorphic, or simply continuous, extension.

As $w \to \infty$, every square-root factor of $S_{T,C}$ tends to one, so

$$S_{T,C}(w) = \frac{1}{w} + O(w^{-2}).$$

The coefficient of $1/w$ in the Cauchy transform Equation (4.4) must be the total mass of $\lambda_{T,C}$. Thus $\lambda_{T,C}$ has mass one.

Finally, fix a point $x \in (\alpha, \beta) \setminus \overline{T}$ in one of the density components. Taking the upper boundary value in (4.4) and applying the Plemelj formula gives

$$S_{T,C,+}(x) = \mathrm{p.\,v.} \int \frac{d\lambda_{T,C}(s)}{x-s} - i\pi\rho_{T,C}(x).$$

For such x, the factor $\chi_{\alpha,\beta,+}(x)$ is purely imaginary, while every factor $\chi_{u_j,v_j,+}(x)$ is real. Hence $S_{T,C,+}(x)$ is purely imaginary, and therefore

$$\mathrm{p.\,v.} \int \frac{d\lambda_{T,C}(s)}{x-s} = 0.$$

The density $\rho_{T,C}$ is C^1 near x and there is no atom at x, so $U_{\lambda_{T,C}}$ is differentiable at x, giving

$$\frac{d}{dx} U_{\lambda_{T,C}}(x) = -\,\mathrm{p.\,v.} \int \frac{d\lambda_{T,C}(s)}{x-s} = 0.$$

Thus the potential is constant on each component of $(\alpha, \beta) \setminus \overline{T}$. Formula (4.5) follows directly from (4.4) on intervals disjoint from the support. □

The next proposition explains why only two values of the dual potential $U_{\lambda_{T,C}}$ will matter to us.

Proposition 4.3. *For $T \subset [\alpha, \beta]$ a finite union of disjoint open intervals, and $C \in [1, -z)$,*

$$\min_{-1 \le x \le 1} U_{\lambda_{T,C}}(x) = \min\{U_{\lambda_{T,C}}(-1), U_{\lambda_{T,C}}(\alpha)\}.$$

As a consequence of the above proposition, to determine that $U_{\lambda_{T,C}} \ge 0$ on $[-1, 1]$, we only need to check if this inequality holds at α and -1.

Proof. For $-1 < x < r$, all square-root factors in (4.2) are positive, $x - z > 0$, $x - r < 0$, and $x + C \ge 0$. Thus $S_{T,C}(x) \le 0$, so $U_{\lambda_{T,C}}$ is nondecreasing by Equation (4.5). For $r < x < \alpha$, every factor in the rational part is positive, so $S_{T,C}(x) > 0$ and $U_{\lambda_{T,C}}$ decreases.

On $(\alpha, \beta) \setminus T$, the potential $U_{\lambda_{T,C}}$ is constant by Proposition 4.2. On a component of T, the outer square-root factor and the factor for that component are both purely imaginary, and their product is negative real; all remaining factors are positive. Hence $S_{T,C} < 0$ there and the potential $U_{\lambda_{T,C}}$ increases. Finally, $\lambda_{T,C}$ has a positive atom at r, so its potential tends to $+\infty$ at r. Since $\beta \ge 1$, the minimum on $[-1, 1]$ is one of the two stated values.

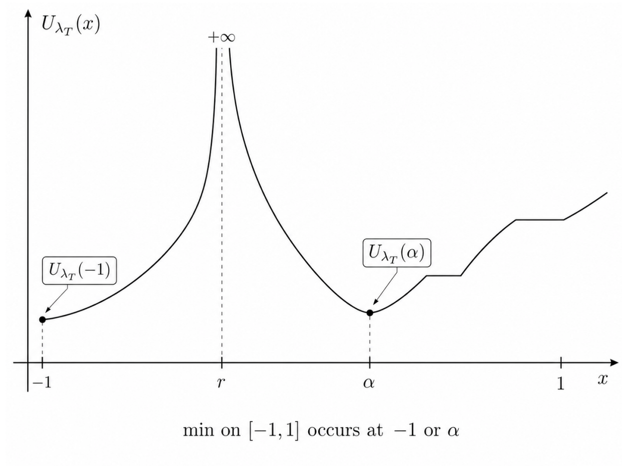

Figure 1. Plot of the dual potential $U_{\lambda_{T,C}}$ on $[-1,1]$.

□

5. Rearranging the exceptional intervals

We continue working with the constants fixed at the beginning of the previous section.

The dual potential $U_{\lambda_{T,C}}$ depends on the arrangement of the components of T, not only on their total length. We shall prove that left compression of intervals in T minimizes its value at -1, while right compression minimizes its value at α. The importance of these values comes from Proposition 4.3.

The use of variational ideas in connection with the Erdős problem 1038 is not new. We refer in particular to [10], which develops a variation framework for certain regular minimizers. Unlike the argument presented here, the variations in [10] are not performed directly at the level of the dual measures.

For a compactly supported probability measure σ, $t \in (\alpha,\beta)$ and $x \in \{-1\} \cup (r,\alpha)$ define

$$K_x^{\sigma}(t) = \frac{1}{2}\int \frac{\log|x-t| - \log|x-s|}{s-t}\, d\sigma(s), \tag{5.1}$$

whenever the integral is finite. At $s=t$ the integrand is defined by continuity to be $1/(x-t)$. Below, $\sigma = \lambda_{T,C}$ so $x \notin \overline{\operatorname{supp} \lambda_{T,C}}$, hence the integral will always be finite.

Lemma 5.1 (Variation of one component)**.** *Suppose that one component (u,v) of T is translated to $(u+\tau, v+\tau)$, with no endpoint collision and with all moving endpoints staying in a compact subinterval of (α,β). If T_τ is the resulting set, then*

$$\frac{d}{d\tau} U_{\lambda_{T_\tau,C}}(x) = K_x^{\lambda_{T_\tau,C}}(v+\tau) - K_x^{\lambda_{T_\tau,C}}(u+\tau), \qquad x \in \{-1\} \cup (r,\alpha). \tag{5.2}$$

Proof. Fix a compact collision-free interval J of τ-values and either $x=-1$ or $x \in (r,\alpha)$. For every $\tau \in J$,

$$\operatorname{supp} \lambda_{T_\tau,C} \subset \{z,r\} \cup [\alpha,\beta], \qquad z < -1 < r < \alpha,$$

so x lies outside the support. We may choose a finite union of positively oriented contours Γ, independent of $\tau \in J$, which encloses $\operatorname{supp} \lambda_{T_\tau,C}$ and the closed moving interval $[u+\tau, v+\tau]$, while leaving x outside every region enclosed by Γ. On a neighborhood of each contour and its interior, choose a holomorphic branch ψ_x whose value on the enclosed real points is

$$\psi_x(s) = \log\frac{1}{|x-s|}.$$

By the Cauchy-transform identity Equation (4.4) and Fubini's theorem we obtain that

$$U_{\lambda_{T_\tau,C}}(x) = \frac{1}{2\pi i}\int_\Gamma \psi_x(w) S_{T_\tau,C}(w)\, dw. \tag{5.3}$$

Indeed, after substituting $S_{T_\tau,C}(w) = \int (w-s)^{-1}\, d\lambda_{T_\tau,C}(s)$, the inner contour integral is $\psi_x(s)$ for every $s \in \operatorname{supp} \lambda_{T_\tau,C}$. Fixing Γ, the above formula in fact holds for any x outside the domains enclosed by Γ.

The contours stay a positive distance from all moving endpoints, so τ-differentiation under the contour integral is legitimate. Only the factor corresponding to $(u+\tau, v+\tau)$ depends on τ, and

$$\partial_\tau S_{T_\tau,C}(w) = \frac{1}{2}\left(\frac{S_{T_\tau,C}(w)}{w-v-\tau} - \frac{S_{T_\tau,C}(w)}{w-u-\tau}\right).$$

Consequently,

$$\frac{d}{d\tau} U_{\lambda_{T_\tau,C}}(x) = \frac{1}{2} I(v+\tau) - \frac{1}{2} I(u+\tau), \tag{5.4}$$

where, for $t \in [u+\tau, v+\tau]$,

$$I(t) = \frac{1}{2\pi i}\int_\Gamma \frac{\psi_x(w) S_{T_\tau,C}(w)}{w-t}\, dw.$$

Again substituting the Cauchy transform from Equation (4.4) and using Fubini gives

$$I(t) = \int \left[\frac{1}{2\pi i}\int_\Gamma \frac{\psi_x(w)}{(w-t)(w-s)}\, dw\right] d\lambda_{T_\tau,C}(s).$$

For $s \neq t$, the inner integral is

$$\frac{\psi_x(s) - \psi_x(t)}{s-t}.$$

For $s = t$, the two simple poles coalesce and the inner integral is $\psi_x'(t) = 1/(x-t) \neq 0$. Hence we obtain that

$$I(t) = \int \frac{\log|x-t| - \log|x-s|}{s-t}\, d\lambda_{T_\tau,C}(s) = 2K_x^{\lambda_{T_\tau,C}}(t), \quad t \in [u+\tau, v+\tau].$$

Substitution into Equation (5.4) proves Equation (5.2). $\square$

The derivative of the kernel can be expressed in terms of the following two functions. They are needed only in the next two results:

$$a(u) = \frac{u(u-1-\log u)}{(u-1)^2}, \qquad b(v) = \frac{v(1+v+\log v)}{(1+v)^2},$$

with the continuous value $a(1) = 1/2$.

Lemma 5.2. *The function a is increasing on $(0,\infty)$, and*

$$\begin{aligned} a(u) &> \frac{1}{3} && \text{for } u \geq \frac{12}{25}, \\ b(v) &\leq \frac{2}{9} && \text{for } 0 < v \leq \frac{1}{2}, \\ a(u) &< b(v) && \text{whenever } 0 < u < v \text{ and } v > 1. \end{aligned}$$

Proof. One computes

$$a'(u) = \frac{(u+1)\log u - 2(u-1)}{(u-1)^3} > 0, \quad u \neq 1,$$

and note that that $a'(1) = 1/6 > 0$. Thus the first bound is equivalent to $a(12/25) \approx 0.380 > 1/3$.

Also, $\log v \leq v - 1$ gives $b(v) \leq 2v^2/(1+v)^2 \leq 2/9$ for $v \leq 1/2$.

Finally,

$$b(v)-a(v)=\frac{2v\big((v^2+1)\log v-v^2+1\big)}{(v-1)^2(v+1)^2}>0 \qquad (v>1).$$

Indeed. For this it is enough to prove that

$$(v^2+1)\log v-v^2+1>0 \qquad (v>1).$$

Put $v=e^y$, where $y>0$, and define

$$F(y):=(e^{2y}+1)y-e^{2y}+1.$$

We have $F(0)=0$ and $F'(y)=e^{2y}+2ye^{2y}+1-2e^{2y}=(2y-1)e^{2y}+1$, so $F'(0)=0$. Differentiating once more,

$$F''(y)=2(2y-1)e^{2y}+2e^{2y}=4ye^{2y}.$$

Thus $F''(y)>0,\ y>0$. It follows that $F(y)>0$ for $y>0$ proving the claim. Since a is increasing, the last assertion follows. ☐

Proposition 5.3 (Kernel monotonicity)**.** *The map $t\mapsto K_{-1}^{\lambda_{T,C}}(t)$ is strictly increasing on (α,β). Moreover, for every $\frac{r+\beta}{2}<x<\alpha$, the function $t\mapsto K_x^{\lambda_{T,C}}(t)$ is strictly decreasing on (α,β).*

Proof. Differentiating the integrand in (5.1), with $\lambda_{T,C}$ fixed, gives

$$\partial_t\frac{\log|x-t|-\log|x-s|}{s-t}=\frac{\xi-1-\log|\xi|}{(t-x)^2(\xi-1)^2},\qquad \xi=\frac{s-x}{t-x}. \tag{5.5}$$

The apparent singularity at $s=t$ is again removable: the quotient in ξ tends to $1/2$ as $\xi\to1$. Thus differentiation under the integral is justified, because x is positive distance away from the support of $\lambda_{T,C}$ and the integrand is jointly continuous in (s,t). We have the following formula relating the above derivative to the functions a and b:

$$\partial_t\frac{\log|x-t|-\log|x-s|}{s-t}=\begin{cases}\dfrac{a\left(\frac{s-x}{t-x}\right)}{(t-x)(s-x)}, & s>x,\\[2ex] -\dfrac{b\left(\frac{x-s}{t-x}\right)}{(t-x)(x-s)}, & s<x.\end{cases}$$

Since

$$K_x^{\lambda}(t)=\frac12\int\frac{\log|x-t|-\log|x-s|}{s-t}\,d\lambda(s),$$

This yields the following useful formula

$$2(t-x)\,\partial_tK_x^{\lambda_{T,C}}(t)=\int_{s>x}a\left(\frac{s-x}{t-x}\right)\frac{d\lambda_{T,C}(s)}{s-x}-\int_{s<x}b\left(\frac{x-s}{t-x}\right)\frac{d\lambda_{T,C}(s)}{x-s}.$$

First take $x=-1$. The part of $\lambda_{T,C}$ supported on $[r,\infty)$ contributes $\xi=(s+1)/(t+1)>0$, while the atom at z gives $\xi=-(-1-z)/(t+1)<0$. If m_z is the mass of that atom, then

$$\partial_tK_{-1}^{\lambda_{T,C}}(t)=\frac{1}{2(t+1)}\left(\int_{s\ge r}a\left(\frac{s+1}{t+1}\right)\frac{d\lambda_{T,C}(s)}{s+1}-m_zb\left(\frac{-1-z}{t+1}\right)\frac{1}{-1-z}\right).$$

For $\alpha<t<\beta$, we have

$$\frac{s+1}{t+1}\ge\frac{1}{2+(D-1.708)/2}\approx0.484>\frac{12}{25},\qquad \frac{-1-z}{t+1}=\frac{D-1-r-h}{t+1}\le\frac{D-1}{1+\alpha}\approx0.462<\frac12.$$

By Lemma 5.2,

$$\partial_tK_{-1}^{\lambda_{T,C}}(t)\ge\frac{1}{2(t+1)}\left(\frac13\int_{s\ge r}\frac{d\lambda_{T,C}(s)}{s+1}-\frac29\frac{m_z}{-1-z}\right).$$

The Cauchy-transform identity Equation (4.4) gives

$$S_{T,C}(-1) = \frac{m_z}{-1-z} - \int_{s\geq r} \frac{d\lambda_{T,C}(s)}{s+1} \leq 0.$$

The estimate follows from (4.2), since $C \geq 1, -1-z > 0, -1-r < 0$. Consequently,

$$\int_{s\geq r} \frac{d\lambda_{T,C}(s)}{s+1} \geq \frac{m_z}{-1-z},$$

and hence

$$\partial_t K_{-1}^{\lambda_{T,C}}(t) \geq \frac{1}{18(t+1)} \frac{m_z}{-1-z} > 0.$$

Thus $K_{-1}^{\lambda_{T,C}}$ is strictly increasing.

Now we deal with the case $\frac{r+\beta}{2} < x < \alpha$. This interval is nonempty: by (6.1), we have that

$$2\alpha - r - \beta = 2\alpha - 1 - r - \frac{h}{2} \geq 2\alpha - 1 - (D - 1.708) \approx 0.482 > 0.$$

The support of $\lambda_{T,C}$ to the right of x lies in $(\alpha, \beta]$, while the only atoms to its left are those at z and r. Writing $m_y = \lambda_{T,C}(\{y\})$ for $y \in \{z, r\}$, equation (5.5) becomes

$$2(t-x)\,\partial_t K_x^{\lambda_{T,C}}(t) = \int_{s>\alpha} a\left(\frac{s-x}{t-x}\right) \frac{d\lambda_{T,C}(s)}{s-x} - \sum_{y\in\{z,r\}} m_y b\left(\frac{x-y}{t-x}\right) \frac{1}{x-y}. \tag{5.6}$$

Here the integral includes the possible atom at β.

For $s \leq \beta$, $y \in \{z, r\}$, and $\alpha < t < \beta$, we have

$$0 < \frac{s-x}{t-x} \leq \frac{\beta-x}{t-x} < \frac{x-r}{t-x} \leq \frac{x-y}{t-x}, \qquad \frac{x-y}{t-x} > 1. \tag{5.7}$$

Indeed, the strict middle inequality is equivalent to $x > (r+\beta)/2$, and the last assertion follows from $t - x < \beta - x < x - r \leq x - y$. Hence Lemma 5.2 gives

$$a\left(\frac{\beta-x}{t-x}\right) < b\left(\frac{x-y}{t-x}\right) \qquad (y = z, r).$$

On the other hand, evaluating the Cauchy-transform identity Equation (4.4) at $x \in (r, \alpha)$ gives

$$S_{T,C}(x) = \sum_{y\in\{z,r\}} \frac{m_y}{x-y} - \int_{s>\alpha} \frac{d\lambda_{T,C}(s)}{s-x} > 0. \tag{5.8}$$

The strict sign follows directly from (4.2): every factor is positive on (r, α). Since a is increasing and positive on $(0, \infty)$, Equations (5.7) and (5.8) imply the strict chain

$$\int_{s>\alpha} a\left(\frac{s-x}{t-x}\right) \frac{d\lambda_{T,C}(s)}{s-x} \leq a\left(\frac{\beta-x}{t-x}\right) \int_{s>\alpha} \frac{d\lambda_{T,C}(s)}{s-x} < a\left(\frac{\beta-x}{t-x}\right) \sum_{y\in\{z,r\}} \frac{m_y}{x-y}$$
$$< \sum_{y\in\{z,r\}} m_y b\left(\frac{x-y}{t-x}\right) \frac{1}{x-y}.$$

By (5.6), this proves $\partial_t K_x^{\lambda_{T,C}}(t) < 0$ for every $t \in (\alpha, \beta)$. ☐

Let

$$T_{\mathrm{L}} := (\alpha, \alpha + h), \qquad T_{\mathrm{R}} := (\beta - h, \beta),$$

with both intervals interpreted as empty when $h = 0$.

Corollary 5.4 (Interval rearrangement monotonicity)**.** *For $T \subset [\alpha, \beta]$ a finite union of disjoint open intervals of total length h, and $C \in [1, -z)$. We have*

$$U_{\lambda_{T,C}}(-1) \geq U_{\lambda_{T_{\mathrm{L}},C}}(-1),$$
$$U_{\lambda_{T,C}}(\alpha) \geq U_{\lambda_{T_{\mathrm{R}},C}}(\alpha).$$

Proof. We first assume that all endpoints of T lie in (α, β).

For the inequality at -1, compress the components successively to the left. During a collision-free stage in which (u, v) is moved to $(u - \tau, v - \tau)$, Lemma 5.1 gives

$$\frac{d}{d\tau} U_{\lambda_{T_\tau,C}}(-1) = K_{-1}^{\lambda_{T_\tau,C}}(u - \tau) - K_{-1}^{\lambda_{T_\tau,C}}(v - \tau) \leq 0,$$

because $K_{-1}^{\lambda_{T_\tau,C}}$ is increasing by Proposition 5.3. At a collision, or when the leftmost component reaches α, continuity follows from Lemma 5.5. After all components have been compressed, we obtain

$$U_{\lambda_{T,C}}(-1) \geq U_{\lambda_{T_{\mathrm{L}},C}}(-1). \tag{5.9}$$

For the right compression, fix $\frac{r+\beta}{2} < x < \alpha$. Move the rightmost component to the right until it reaches β or meets the next component, merge colliding components, and continue. During every collision-free stage, Lemma 5.1 and Proposition 5.3 give

$$\frac{d}{d\tau} U_{\lambda_{T_\tau,C}}(x) = K_x^{\lambda_{T_\tau,C}}(v + \tau) - K_x^{\lambda_{T_\tau,C}}(u + \tau) \leq 0.$$

Using Lemma 5.5 at collisions and when the rightmost component reaches β, the completed compression gives

$$U_{\lambda_{T,C}}(x) \geq U_{\lambda_{T_{\mathrm{R}},C}}(x), \qquad \frac{r+\beta}{2} < x < \alpha. \tag{5.10}$$

We now remove the assumption that all endpoints of T lie in the interior. Choose finite unions $T_n \subset (\alpha, \beta)$, each of total length h, whose endpoints converge to those of T. Such an approximation is possible because

$$h < \beta - \alpha,$$

which follows from $h \leq r + h \leq D - 1.708 \approx 0.126 < 0.391 \approx 2(1 - \alpha)$ (6.1). Applying (5.9) and (5.10) to T_n, and then using Lemma 5.5 at the fixed points -1 and $x \in ((r + \beta)/2, \alpha)$, gives

$$U_{\lambda_{T,C}}(-1) \geq U_{\lambda_{T_{\mathrm{L}},C}}(-1) \quad \text{and} \quad U_{\lambda_{T,C}}(x) \geq U_{\lambda_{T_{\mathrm{R}},C}}(x) \qquad \frac{r+\beta}{2} < x < \alpha.$$

Finally, let $x \nearrow \alpha$. Neither $\lambda_{T,C}$ nor $\lambda_{T_{\mathrm{R}},C}$ has an atom at α. By (4.3), each density is either zero on a right-neighborhood of α or is $O(\sqrt{s - \alpha})$ there. Hence both potentials $U_{\lambda_{T,C}}$, $U_{\lambda_{T_{\mathrm{R}},C}}$ are continuous at α. Passing to the limit in the last inequality yields

$$U_{\lambda_{T,C}}(\alpha) \geq U_{\lambda_{T_{\mathrm{R}},C}}(\alpha),$$

which proves the second assertion. ☐

Lemma 5.5 (Continuity under endpoint motion)**.** *Fix r, h, C as described at the beginning of Section 4. Let the endpoints of the components of $T_n \subset [\alpha, \beta]$ converge to those of T, allowing components to collide, disappear, or meet α or β, as specified below. Then*

$$U_{\lambda_{T_n,C}}(x) \to U_{\lambda_{T,C}}(x)$$

for every fixed $x = -1$ or $x \in (r, \alpha)$.

If an interval (u_n, v_n) collapses to a point, so that $u_n, v_n \to c$, then $\chi_{u_n,v_n}(w) = \sqrt{\frac{w-u_n}{w-v_n}} \to 1$ uniformly on compact sets disjoint from c. Likewise, if two adjacent intervals collide at a point c, then $\chi_{u,c}(w)\chi_{c,v}(w) = \sqrt{\frac{w-u}{w-c}}\sqrt{\frac{w-c}{w-v}} = \sqrt{\frac{w-u}{w-v}} = \chi_{u,v}(w)$. Here all square roots are taken with the normalized branches used above, namely the branches that tend to 1 as $w \to \infty$.

Proof. Choose a finite union of contours Γ, independent of n, which encloses the supports of all the measures $\lambda_{T_n,C}$ and of $\lambda_{T,C}$, but does not enclose x. By the same Cauchy-transform and contour argument used to obtain (5.3),

$$U_{\lambda_{T_n,C}}(x) = \frac{1}{2\pi i}\int_\Gamma \psi_x(w) S_{T_n,C}(w)\,dw.$$

As the interval endpoints converge, the transforms $S_{T_n,C}$ converge u
niformly to $S_{T,C}$ on the fixed contour Γ. Therefore

$$U_{\lambda_{T_n,C}}(x) \to \frac{1}{2\pi i}\int_\Gamma \psi_x(w) S_{T,C}(w)\,dw = U_{\lambda_{T,C}}(x).$$

□

6. The endpoint certificate and the lower bound

We remind the readers of the list of our constants and the constraints imposed upon them:

$$r \geq 0, \qquad h \geq 0, \qquad r + h \leq D - 1.708, \tag{6.1}$$

and put

$$\beta := 1 + \frac{h}{2}, \qquad z := r + h - D.$$

We also have the chain of estimates $z < \ell < -1 < 0 < r < \alpha < 1 < \beta$, and we will work with an open set T:

$$T = \bigcup_{j=1}^{m} (u_j, v_j) \subset [\alpha, \beta]$$

Recall also that

$$D = 1.834430475762661711090\ldots, \qquad \alpha = 0.804461754260998666\ldots.$$

For the two compressed intervals

$$T_{\mathrm{R}} := (\beta - h, \beta), \qquad T_{\mathrm{L}} := (\alpha, \alpha + h),$$

let $\mathsf{R}(r,h,C)$ and $\mathsf{L}(r,h,C)$ denote, respectively, the right-hand sides of Equations (B.4) and (B.5), that are defined for $r, h \in [0, 0.127]$, $C \in \mathbb{R}$. For fixed r, h, both are real-valued affine functions of C for every $C \in \mathbb{R}$. When $1 \leq C < -z$, the dual measures $\lambda_{T,C}$ are defined and according to (B.6) we have

$$\mathsf{R}(r,h,C) = U_{\lambda_{T_{\mathrm{R}},C}}(\alpha), \qquad \mathsf{L}(r,h,C) = U_{\lambda_{T_{\mathrm{L}},C}}(-1). \tag{6.2}$$

The function R has positive C-derivative for all r, h satisfying $r + h \leq D - 1.708 < 0.127$, by (6.5) below. We therefore define $C_0(r,h) \in \mathbb{R}$ to be the unique zero of $\mathsf{R}(r,h,C_0(r,h)) = 0$:

$$C_0(r,h) := -\frac{\mathsf{R}(r,h,0)}{\partial_C \mathsf{R}(r,h,0)}. \tag{6.3}$$

The point of introducing R and L separately from the dual measures is that $C_0(r,h)$ is defined algebraically even when it does not lie in the admissible interval $[1, -z)$. Only the final value of C chosen in Proposition 6.2 will be required to belong to that interval.

Lemma 6.1. *On the square $0 \le r, h \le 0.127$, outward-rounded Arb arithmetic certifies*

$$\frac{\partial}{\partial h}\mathsf{L}(r,h,C_0(r,h)) > 0, \qquad \frac{\partial^2}{\partial r^2}\mathsf{L}(r,0,C_0(r,0)) > 0, \tag{6.4}$$

$$\frac{\partial}{\partial C}\mathsf{R}(r,h,C) > 0, \qquad (D-r-h)-C_0(r,h) > 0. \tag{6.5}$$

When $C = 1$ we have the following estimates:

$$\frac{\partial}{\partial r}\mathsf{L}(r,h,1) < 0, \qquad \left(\frac{\partial}{\partial h}-\frac{\partial}{\partial r}\right)\mathsf{L}(r,h,1) > 0, \tag{6.6}$$

$$\mathsf{L}(D-1.708,0,1) > 0. \tag{6.7}$$

Proof. The inequalities are proved by `erdos1038_exact_endpoint_certificate.py`, using second-order automatic-differentiation jets, rational box subdivision, interval Taylor bounds, and 384-bit Arb balls. The program imports the rigorous enclosures of α and D produced by the earlier used script `contact_existence.py`. □

Proposition 6.2. *For every*

$$r \ge 0, \qquad h \ge 0, \qquad r+h \le D-1.708,$$

there is a number C with

$$1 \le C < -z = -(r+h-D)$$

such that

$$U_{\lambda_{T_{\mathrm{L}},C}}(-1) \ge 0, \qquad U_{\lambda_{T_{\mathrm{R}},C}}(\alpha) \ge 0.$$

Proof. By Lemma B.1 and the strict convexity in (6.4), the function

$$r \longmapsto \mathsf{L}(r,0,C_0(r,0))$$

has minimum zero at $r = r_*$, where r_* is defined under Definition A.2. The positive derivative with respect to h in (6.4) gives

$$\mathsf{L}(r,h,C_0(r,h)) \ge \mathsf{L}(r,0,C_0(r,0)) \ge 0. \tag{6.8}$$

Suppose first that $C_0(r,h) > 1$, and choose $C = C_0(r,h)$. The second inequality in (6.5) gives $C < -z$, so this value of C is admissible ($C \in [1,-z]$). Therefore (6.2), (6.3), and (6.8) give

$$U_{\lambda_{T_{\mathrm{R}},C}}(\alpha) = \mathsf{R}(r,h,C) = 0, \qquad U_{\lambda_{T_{\mathrm{L}},C}}(-1) = \mathsf{L}(r,h,C) \ge 0.$$

Now suppose that $C_0(r,h) \le 1$. Along the segment $(r+h-\tau,\tau)$, $0 \le \tau \le h$, the second inequality in (6.6) gives the first inequality below; the first inequality in (6.6), the bound $r+h \le D-1.708$, and (6.7) give the rest:

$$\mathsf{L}(r,h,1) \ge \mathsf{L}(r+h,0,1) \ge \mathsf{L}(D-1.708,0,1) > 0.$$

Because $\mathsf{R}(r,h,C)$ is affine in C, has positive slope, and vanishes at $C = C_0(r,h)$,

$$\mathsf{R}(r,h,1) = \frac{\partial}{\partial C}\mathsf{R}(r,h,C)\big(1-C_0(r,h)\big) \ge 0.$$

The choice $C = 1$ is admissible because $-z \ge 1.708 > 1$. Hence (6.2) gives

$$U_{\lambda_{T_{\mathrm{L}},1}}(-1) = \mathsf{L}(r,h,1) > 0, \qquad U_{\lambda_{T_{\mathrm{R}},1}}(\alpha) = \mathsf{R}(r,h,1) \ge 0.$$

Thus $C = 1$ works in this case. □

Finally we are ready to prove that for any configuration of intervals $T \subset [\alpha,\beta]$ there exists C such that $U_{\lambda_{T,C}}(x) \ge 0$:

Corollary 6.3. *Let $T \subset [\alpha,\beta]$ be a finite union of open intervals of total length h, and choose C as in Proposition 6.2. Then*

$$U_{\lambda_{T,C}}(x) \ge 0 \qquad (-1 \le x \le 1).$$

Proof. By Corollary 5.4 and Proposition 6.2,

$$U_{\lambda_{T,C}}(-1) \ge U_{\lambda_{T_{\mathrm{L}},C}}(-1) \ge 0, \qquad U_{\lambda_{T,C}}(\alpha) \ge U_{\lambda_{T_{\mathrm{R}},C}}(\alpha) \ge 0.$$

An application of Proposition 4.3 concludes the proof. □

Proof of the lower bound in Theorem 1.1. Assume for contradiction that μ is a finite atomic counterexample, i.e., $\mathcal{L}_\mu < D$. Put it in the strengthened normal form and choose h, β, T as in Lemma 3.3. Thus

$$T = \operatorname{int}(E_\mu \cap [\alpha, \beta]), \qquad |T| = h, \qquad \beta \notin E_\mu, \qquad z = r + h - D < \ell.$$

Since -1 is the leftmost atom, U_μ is strictly increasing on $(-\infty, -1)$. The point ℓ is the left endpoint of the positive component containing -1. Since $z < \ell$,

$$U_\mu(z) < 0.$$

Choose C as in Proposition 6.2, and set $\lambda = \lambda_{T,C}$, according to Definition 4.1.

The measure λ is a probability measure. It has positive atoms at z and r, and it may also have an atom at β if a component of T has right endpoint β. We check the sign of U_μ on its support. The atom at z gives a strictly negative contribution. The atom at r gives zero contribution: r is an endpoint of (ℓ, r), it is not an atom of μ, and continuity gives $U_\mu(r) = 0$. The possible atom at β gives a nonpositive contribution because $\beta \notin E_\mu$. Finally, the absolutely continuous part of λ is supported, up to finitely many endpoints, on $[\alpha, \beta] \setminus T$, where $U_\mu \le 0$. Therefore

$$\int_{\mathbb{R}} U_\mu \, d\lambda < 0.$$

On the other hand, every atom of μ lies in $[-1, 1]$, and Corollary 6.3 gives

$$U_\lambda(x) \ge 0 \qquad (-1 \le x \le 1).$$

Thus $U_\lambda \ge 0$ μ-almost everywhere, where the value $+\infty$ is allowed. By Lemma 2.4,

$$\int_{\mathbb{R}} U_\mu \, d\lambda = \int_{\mathbb{R}} U_\lambda \, d\mu \ge 0,$$

contradicting the preceding strict inequality. Hence no finite atomic counterexample exists.

From Corollary 2.3 we obtain that

$$\inf_f \mathcal{L}_f = \inf_\mu \mathcal{L}_\mu \ge D,$$

where f runs over non-constant monic polynomials with all roots in $[-1, 1]$, and μ runs over Borel propbability measures supported in $[-1, 1]$. □

7. Sharpness of the lower bound

The preceding sections prove that every finite atomic probability measure under consideration satisfies $\mathcal{L}_\mu \ge D$. The purpose of this section is to show that this inequality is sharp, thus finishing the proof of Theorem 1.1.

We first construct a probability measure μ_* on $[-1, 1]$ for which E_{μ_*} is an interval of length exactly D. This measure is the same as that constructed by Tao in [11, Section 4], but we approach it slightly differently here, to fit our context better.

We use the following functions (coordinates) in our calculations below. For $0 < y < 1$ and $0 < q < 1$, put

$$\Lambda(y) := \log \frac{1 + y}{1 - y}, \qquad X_q(y) := \frac{2q^2 - 1 - y^2}{1 - y^2}. \tag{7.1}$$

For fixed q, the map X_q is strictly decreasing, since

$$X_q'(y) = -\frac{4y(1 - q^2)}{(1 - y^2)^2} < 0.$$

It satisfies

$$X_q(0) = 2q^2 - 1, \qquad X_q(q) = -1, \qquad X_q(y) \to -\infty, \quad y \nearrow 1,$$

and its inverse on $(-\infty, 2q^2 - 1)$ is

$$y = \sqrt{\frac{2q^2 - 1 - x}{1 - x}}.$$

Define

$$p(q) := \frac{\log(2/(1 - q^2))}{\Lambda(q)}. \tag{7.2}$$

The significance of this number is that the optimal measure μ below will have mass $p(q)$ at the atom -1.

Appendix A defines a box $\mathcal{B}_* \subset \mathbb{R}^3$ in (A.9) and a three-component map $\mathcal{F}$ in (A.8). By Lemma A.1, the system has at least one solution in $\mathcal{B}_*$. Fix one such solution and denote it by

$$(q_*, u_*, v_*) \in \mathcal{B}_*, \qquad \mathcal{F}(q_*, u_*, v_*) = 0. \tag{7.3}$$

Put

$$p_* := p(q_*).$$

The certified enclosures in Appendix A imply

$$0 < p_* < q_* < 1, \qquad 0 < u_* < q_* < v_* < 1.$$

The first two components of $\mathcal{F}(q_*, u_*, v_*) = 0$ are

$$\Lambda(u_*) = p_* \log \frac{q_* + u_*}{q_* - u_*}, \tag{7.4}$$

$$\Lambda(v_*) = p_* \log \frac{q_* + v_*}{v_* - q_*}. \tag{7.5}$$

Finally, we introduce the following related constants, whose numeric values can be found under (A.17):

$$\alpha = 2q_*^2 - 1, \qquad r_* := X_{q_*}(u_*), \qquad z_* := X_{q_*}(v_*), \qquad D = r_* - z_*. \tag{7.6}$$

Thus

$$z_* < -1 < r_* < \alpha.$$

The construction in this section uses only (7.4) and (7.5), third equation of $\mathcal{F}(q_*, u_*, v_*) = 0$ is not needed here.

Define

$$\mu_* := p_* \delta_{-1} + \frac{s + 1 - 2p_* q_*}{\pi(s+1)\sqrt{(s - \alpha)(1 - s)}} \mathbf{1}_{(\alpha,1)}(s)\, ds. \tag{7.7}$$

The density is positive because, for $\alpha \le s \le 1$,

$$s + 1 - 2p_* q_* \ge \alpha + 1 - 2p_* q_* = 2q_*(q_* - p_*) > 0.$$

Lemma 7.1 (The optimal measure)**.** *The measure μ_* is a probability measure. Its Cauchy transform is*

$$\int_{\mathbb{R}} \frac{d\mu_*(s)}{w - s} = \frac{w + 1 - 2p_* q_*}{(w+1)\sqrt{(w - \alpha)(w - 1)}}, \tag{7.8}$$

where the square root is the branch asymptotic to w at infinity. Its potential U_{μ^*} vanishes on $[\alpha, 1]$. For $x < \alpha$, write $x = X_{q_*}(y)$, equivalently

$$y = \sqrt{\frac{\alpha - x}{1 - x}}, \qquad 0 < y < 1.$$

Then, for $y \neq q_*$,

$$U_{\mu_*}(X_{q_*}(y)) = -\Lambda(y) + p_* \log\left|\frac{q_*+y}{q_*-y}\right|. \tag{7.9}$$

Proof. Set

$$G(w) := \frac{w+1-2p_*q_*}{(w+1)\sqrt{(w-\alpha)(w-1)}},$$

where the square root is the branch asymptotic to w at infinity. We verify (7.8) by the same slit-contour argument used in the proof of Proposition 4.2. The required ingredients are as follows.

(i) The function G is holomorphic on $\mathbb{C}\setminus(\{-1\}\cup[\alpha,1])$, with a simple pole at -1. Since $1+\alpha = 2q_*^2$, the chosen square-root branch satisfies

$$\sqrt{(-1-\alpha)(-2)} = -2q_*,$$

and hence

$$\operatorname{Res}_{w=-1} G(w) = \frac{-2p_*q_*}{-2q_*} = p_*.$$

(ii) For $\alpha < s < 1$, the upper and lower boundary values of the square root are

$$\sqrt{(s-\alpha)(s-1)}_+ = i\sqrt{(s-\alpha)(1-s)}, \qquad \sqrt{(s-\alpha)(s-1)}_- = -i\sqrt{(s-\alpha)(1-s)}.$$

Consequently,

$$G_+(s) - G_-(s) = -2\pi i\,\frac{s+1-2p_*q_*}{\pi(s+1)\sqrt{(s-\alpha)(1-s)}},$$

which is exactly $-2\pi i$ times the density in (7.7).

(iii) At α and 1, the function G has only square-root singularities. Thus the integrals over small circles around these endpoints tend to zero in the slit-contour argument.

(iv) Finally,

$$G(w) = \frac{1}{w} + O(w^{-2}) \qquad (w\to\infty).$$

Applying the residue theorem on the plane slit along $[\alpha,1]$, exactly as in the proof of Proposition 4.2, the residue at -1 and the jump across the cut $[\alpha,1]$ give

$$G(w) = \frac{p_*}{w+1} + \int_\alpha^1 \frac{1}{w-s}\,\frac{s+1-2p_*q_*}{\pi(s+1)\sqrt{(s-\alpha)(1-s)}}\,ds = \int_{\mathbb{R}} \frac{d\mu_*(s)}{w-s}.$$

The coefficient of $1/w$ in this Cauchy transform is the total mass of μ_*; by (iv), that coefficient is one. Since the atom and the density are positive, μ_* is a probability measure. This proves (7.8).

For $x \in (\alpha,1)$, the boundary value of the square root in (7.8) is purely imaginary. The real part of the boundary value of the Cauchy transform is therefore zero. The Plemelj argument from Proposition 4.2 gives

$$U'_{\mu_*}(x) = -\,\mathrm{p.\,v.}\int \frac{d\mu_*(s)}{x-s} = 0 \qquad (\alpha < x < 1).$$

Thus the potential is constant on $(\alpha,1)$, and continuity extends this constant to $[\alpha,1]$.

We compute both this constant and the potential to the left of the cut. For $x<\alpha$, the branch in (7.8) is

$$\sqrt{(x-\alpha)(x-1)} = -\sqrt{(\alpha-x)(1-x)}.$$

Put $x = X_{q_*}(y)$. Direct calculation gives

$$1 - X_{q_*}(y) = \frac{2(1-q_*^2)}{1-y^2}, \qquad X_{q_*}(y)+1 = \frac{2(q_*^2-y^2)}{1-y^2},$$

$$\sqrt{(X_{q_*}(y)-\alpha)(X_{q_*}(y)-1)} = -\frac{2y(1-q_*^2)}{1-y^2}, \qquad\qquad X'_{q_*}(y) = -\frac{4y(1-q_*^2)}{(1-y^2)^2}.$$

For $y \neq q_*$, the point $X_{q_*}(y)$ lies outside $\operatorname{supp}\mu_*$, so (7.8) and the chain rule give

$$\begin{aligned}\frac{d}{dy}U_{\mu_*}(X_{q_*}(y)) &= U'_{\mu_*}(X_{q_*}(y))X'_{q_*}(y) = -G(X_{q_*}(y))X'_{q_*}(y)\\ &= -\frac{\dfrac{2(q_*^2-y^2)-2p_*q_*(1-y^2)}{1-y^2}}{\dfrac{2(q_*^2-y^2)}{1-y^2}\left(-\dfrac{2y(1-q_*^2)}{1-y^2}\right)}\left(-\frac{4y(1-q_*^2)}{(1-y^2)^2}\right)\\ &= -\frac{2}{1-y^2}+\frac{2p_*q_*}{q_*^2-y^2}.\end{aligned}$$

On the other hand,

$$\Lambda'(y) = \frac{2}{1-y^2}, \qquad \frac{d}{dy}\log\left|\frac{q_*+y}{q_*-y}\right| = \frac{1}{q_*+y}+\frac{1}{q_*-y} = \frac{2q_*}{q_*^2-y^2}.$$

Therefore

$$\frac{d}{dy}U_{\mu_*}(X_{q_*}(y)) = -\Lambda'(y) + p_*\frac{d}{dy}\log\left|\frac{q_*+y}{q_*-y}\right|,$$

and hence

$$U_{\mu_*}(X_{q_*}(y)) = c - \Lambda(y) + p_*\log\left|\frac{q_*+y}{q_*-y}\right|. \tag{7.10}$$

The same constant occurs on both sides of $y = q_*$: the continuous part of the potential is continuous at -1, and the logarithmic singularity of the atom is the same from the left and the right.

As $y \searrow 0$, one has $X_{q_*}(y) \nearrow \alpha$, and both logarithmic terms in (7.10) tend to zero. Thus c is the constant value of the potential on $[\alpha, 1]$. To determine it, use

$$U_{\mu_*}(x) = -\log|x| + o(1) \qquad (x \to -\infty).$$

As $y \nearrow 1$,

$$-X_{q_*}(y) = \frac{1-q_*^2}{1-y}(1+o(1)), \qquad \Lambda(y) = \log\frac{2}{1-y} + o(1),$$

and

$$\log\frac{q_*+y}{y-q_*} = \Lambda(q_*) + o(1).$$

Substitution into (7.10) gives

$$c = \log\frac{2}{1-q_*^2} - p_*\Lambda(q_*) = 0$$

by the definition of p_*. This proves both the vanishing on $[\alpha, 1]$ and (7.9). □

Lemma 7.2 (The positive component)**.** *The positive set of μ_* is $E_{\mu_*} = (z_*, r_*)$. Consequently,*

$$\mathcal{L}(\mu_*) = r_* - z_* = D.$$

Proof. On $(0, q_*)$, differentiation of the right side of (7.9) gives

$$U_{\mu_*}(X_{q_*}(y))' = -\frac{2}{1-y^2} + \frac{2p_*q_*}{q_*^2-y^2}. \tag{7.11}$$

It vanishes exactly when

$$y^2 = \frac{q_*(q_*-p_*)}{1-p_*q_*}.$$

Because $0 < p_* < q_* < 1$, this gives exactly one critical point in $(0, q_*)$. The derivative is negative at $y = 0$ and tends to $+\infty$ as $y \nearrow q_*$. The potential equals zero at $y = 0$, first decreases, and then increases to $+\infty$. It therefore has exactly one nonzero zero in $(0, q_*)$. By (7.4) and (7.9), this zero is u_*. Hence the potential is positive exactly for

$$u_* < y < q_*.$$

On $(q_*, 1)$, the derivative is

$$-\frac{2}{1-y^2} - \frac{2p_*q_*}{y^2 - q_*^2} < 0.$$

The potential tends to $+\infty$ as $y \searrow q_*$ and to $-\infty$ as $y \nearrow 1$. It has exactly one zero on this interval, and (7.5) identifies it as v_*. Thus the potential is positive exactly for

$$q_* < y < v_*.$$

Since X_{q_*} is strictly decreasing and $X_{q_*}(q_*) = -1$, these two ranges give

$$\{x < \alpha : U_{\mu_*}(x) > 0\} = (X_{q_*}(v_*), X_{q_*}(u_*)) = (z_*, r_*).$$

There are no additional positive points. The potential is zero on $[\alpha, 1]$ by Lemma 7.1; for $x > 1$, the Cauchy transform in (7.8) is positive, so $U'_{\mu_*}(x) < 0$. This proves the assertion. ☐

Proof of the sharpness of the lower bound in Theorem 1.1. By Lemma 7.2,

$$\mathcal{L}(\mu_*) = D.$$

By the proof at the and the previous section we know that $\inf_\mu \mathcal{L}_\mu \geq D$, hence $\inf_\mu \mathcal{L}_\mu = D$. By Corollary 2.3 we obtain that $\inf_f \mathcal{L}_f = D$, thus finishing the proof of Theorem 1.1. ☐

Appendix A. Definition and existence of extremal constants

This appendix gives the exact definition of many of the constants (extremal or otherwise) used throughout the paper. Recall Λ, X_q, and $p(q)$ from Equations (7.1) and (7.2):

$$\Lambda(y) := \log\frac{1+y}{1-y}, \qquad X_q(y) := \frac{2q^2 - 1 - y^2}{1 - y^2}, \qquad p(q) := \frac{\log(2/(1-q^2))}{\Lambda(q)},$$

where $0 < y < 1$ and $0 < q < 1$.

For $0 < u < q < v < 1$, put

$$r := X_q(u), \qquad z := X_q(v),$$

and define the two functions:

$$F_{\mathrm{R}}(q, u) := \Lambda(u) - p(q)\log\frac{q+u}{q-u}, \tag{A.1}$$

$$F_{\mathrm{L}}(q, v) := \Lambda(v) - p(q)\log\frac{q+v}{v-q}. \tag{A.2}$$

Since $y \mapsto y\Lambda(y)$ is strictly increasing on $(0, 1)$, the quantity

$$C(q, u, v) := \frac{ru\Lambda(u) - zv\Lambda(v) - (r-z)p(q)\Lambda(q)}{v\Lambda(v) - u\Lambda(u)} \tag{A.3}$$

is well defined. We also use

$$\partial_u F_{\mathrm{R}}(q, u) = \frac{2}{1-u^2} - \frac{2p(q)q}{q^2 - u^2}, \tag{A.4}$$

$$\partial_v F_{\mathrm{L}}(q, v) = \frac{2}{1-v^2} + \frac{2p(q)q}{v^2 - q^2}, \tag{A.5}$$

$$X'_q(y) = -\frac{4y(1-q^2)}{(1-y^2)^2}. \tag{A.6}$$

Define another function by

$$K(q,u,v) := \frac{v(z + C(q,u,v))\partial_v F_{\mathrm{L}}(q,v)}{X_q'(v)} - \frac{u(r + C(q,u,v))\partial_u F_{\mathrm{R}}(q,u)}{X_q'(u)}, \tag{A.7}$$

and combine the three components into

$$\mathcal{F}(q,u,v) := \begin{pmatrix} F_{\mathrm{R}}(q,u) \\ F_{\mathrm{L}}(q,v) \\ K(q,u,v) \end{pmatrix}. \tag{A.8}$$

Let

$$\begin{aligned} q_0 &:= 0.94985834582347031848349802544031979, \\ u_0 &:= 0.89396612207545611611103128550805002, \\ v_0 &:= 0.96455509699582140639698562612728156, \end{aligned}$$

and we set up the box

$$\mathcal{B}_* := [q_0 - 10^{-30}, q_0 + 10^{-30}] \times [u_0 - 10^{-30}, u_0 + 10^{-30}] \times [v_0 - 10^{-30}, v_0 + 10^{-30}]. \tag{A.9}$$

Every vector (q,u,v) of this box satisfies $0 < u < q < v < 1$.

Lemma A.1 (Existence of the parameter triple)**.** *There exists a point $(q,u,v) \in \mathcal{B}_*$ satisfying*

$$\mathcal{F}(q,u,v) = 0. \tag{A.10}$$

Proof. We reduce the three equations to a scalar intermediate-value argument. Put

$$q_- := q_0 - 10^{-31}, \qquad q_+ := q_0 + 10^{-31}, \qquad I_* := [q_-, q_+]. \tag{A.11}$$

The program `contact_existence.py`, using outward-rounded 384-bit Arb arithmetic, implements the scalar argument below and first certifies that

$$0 < p(q) < q < 1 \qquad (q \in I_*), \tag{A.12}$$

and that, uniformly for $q \in I_*$,

$$F_{\mathrm{R}}(q, u_0 - 10^{-30}) > 0 > F_{\mathrm{R}}(q, u_0 + 10^{-30}), \tag{A.13}$$

$$F_{\mathrm{L}}(q, v_0 - 10^{-30}) < 0 < F_{\mathrm{L}}(q, v_0 + 10^{-30}). \tag{A.14}$$

Fix $q \in I_*$. From (A.4),

$$\partial_u F_{\mathrm{R}}(q,u) = \frac{2\big(q(q - p(q)) - (1 - p(q)q)u^2\big)}{(1-u^2)(q^2-u^2)}. \tag{A.15}$$

By (A.12), the numerator in (A.15) has exactly one zero in $(0,q)$, namely

$$u_c(q) := \sqrt{\frac{q(q-p(q))}{1-p(q)q}},$$

and $0 < u_c(q) < q$; indeed,

$$q^2(1-p(q)q) - q(q-p(q)) = p(q)q(1-q^2) > 0.$$

Hence $F_{\mathrm{R}}(q,\cdot)$ is strictly increasing on $(0, u_c(q))$ and strictly decreasing on $(u_c(q), q)$. Since

$$F_{\mathrm{R}}(q,0) = 0, \qquad F_{\mathrm{R}}(q,u) \to -\infty \quad (u \nearrow q),$$

there is exactly one nonzero root $u(q) \in (0,q)$. The signs in (A.13) locate this root in

$$u_0 - 10^{-30} < u(q) < u_0 + 10^{-30}.$$

Moreover, $u(q) > u_c(q)$, so $\partial_u F_{\mathrm{R}}(q, u(q)) < 0$.

Similarly, (A.5) gives

$$\partial_v F_{\mathrm{L}}(q,v) > 0 \qquad (q < v < 1).$$

Together with

$$F_{\mathrm{L}}(q,v)\to-\infty\quad (v\searrow q),\qquad F_{\mathrm{L}}(q,v)\to+\infty\quad (v\nearrow 1),$$

this shows that there is exactly one root $v(q)\in(q,1)$. By (A.14),

$$v_0-10^{-30}<v(q)<v_0+10^{-30}.$$

The implicit-function theorem now shows that $u(q)$ and $v(q)$ are continuous, in fact continuously differentiable, on I_*.

Define the continuous scalar function

$$\kappa(q):=K(q,u(q),v(q)),\qquad q\in I_*.$$

At each endpoint $q_\pm$, the program encloses $u(q_\pm)$ and $v(q_\pm)$ by certified interval bisection and then evaluates K on those enclosures. It certifies the strict signs

$$\kappa(q_-)>0,\qquad \kappa(q_+)<0. \tag{A.16}$$

The intermediate value theorem therefore gives a point $q_*\in(q_-,q_+)$ with $\kappa(q_*)=0$. Setting

$$u_*:=u(q_*),\qquad v_*:=v(q_*),$$

we obtain

$$F_{\mathrm{R}}(q_*,u_*)=F_{\mathrm{L}}(q_*,v_*)=K(q_*,u_*,v_*)=0.$$

The bounds above place (q_*,u_*,v_*) in $\mathcal{B}_*$, proving the claim. The same program exports rigorous enclosures for p_*,α,r_*,z_*, and D; the endpoint certificate imports these enclosures directly. ☐

Definition A.2. Fix one point $(q_*,u_*,v_*)\in\mathcal{B}_*$ supplied by Lemma A.1, and define

$$p_*:=p(q_*),\qquad \alpha:=2q_*^2-1,\qquad r_*:=X_{q_*}(u_*),\qquad z_*:=X_{q_*}(v_*),\qquad D:=r_*-z_*. \tag{A.17}$$

These choices and equations give the exact definitions of the constants below, used throughout the paper.

The certified calculations provide the following approximate values for the constants used in this manuscript:

$$\begin{aligned}
q_*&=0.949858345823470318\ldots, & u_*&=0.893966122075456116\ldots,\\
v_*&=0.964555096995821406\ldots, & p_*&=0.824521729252520379\ldots,\\
\alpha&=0.804461754260998666\ldots, & r_*&=0.0263231076477328506\ldots,\\
z_*&=-1.808107368114928860\ldots, & D&=1.834430475762661711\ldots.
\end{aligned}$$

Appendix B. Explicit formulas for $U_{\lambda_{T_{\mathrm{R}},C}}(\alpha), U_{\lambda_{T_{\mathrm{L}},C}}(-1)$ and symbolic checks

The main point of this appendix is to derive explicit formulas or the integral expressions $U_{\lambda_{T_{\mathrm{R}},C}}(\alpha), U_{\lambda_{T_{\mathrm{L}},C}}(-1)$ and verify them using symbolic calculus Sympy scripts.

For $p<a<b$, define

$$J_{a,b}(p):=\frac{a+b-2p+2\sqrt{(a-p)(b-p)}}{b-a}. \tag{B.1}$$

Then $J_{a,b}(p)>1$ and

$$\frac{d}{dp}\log J_{a,b}(p)=-\frac{1}{\sqrt{(a-p)(b-p)}}. \tag{B.2}$$

For $0\le y<1$, put

$$\Phi(y):=\begin{cases}\dfrac{\operatorname{artanh}\sqrt{y}}{\sqrt{y}}, & y>0,\\ 1, & y=0.\end{cases}=\sum_{n=0}^{\infty}\frac{y^n}{2n+1}. \tag{B.3}$$

Recall that $\beta = 1 + h/2$ and $z = r + h - D$.

We now define the following two real-valued functions for every $C \in \mathbb{R}$ and $r, h \in [0, 0.127]$:

$$\begin{aligned} \mathsf{R}(r,h,C) := & -\log \frac{\beta - h - \alpha}{4} \\ & + \frac{(z+C)\sqrt{(\alpha - z)(\beta - h - z)}}{(z-r)(z-\beta)} \log J_{\alpha,\beta-h}(z) \\ & + \frac{(r+C)\sqrt{(\alpha - r)(\beta - h - r)}}{(r-z)(r-\beta)} \log J_{\alpha,\beta-h}(r) \\ & - \frac{2h(\beta + C)}{(\beta - z)(\beta - r)} \Phi\left(\frac{h}{\beta - \alpha}\right), \end{aligned} \tag{B.4}$$

and

$$\begin{aligned} \mathsf{L}(r,h,C) := & -\log\left(\frac{\beta - \alpha - h}{4} J_{\alpha+h,\beta}(-1)\right) \\ & + \frac{(z+C)(z-\alpha)}{(z-r)\sqrt{(\alpha + h - z)(\beta - z)}} \log \left| \frac{J_{\alpha+h,\beta}(z) - J_{\alpha+h,\beta}(-1)}{J_{\alpha+h,\beta}(-1) - J_{\alpha+h,\beta}(z)^{-1}} \right| \\ & + \frac{(r+C)(r-\alpha)}{(r-z)\sqrt{(\alpha + h - r)(\beta - r)}} \log \left| \frac{J_{\alpha+h,\beta}(r) - J_{\alpha+h,\beta}(-1)}{J_{\alpha+h,\beta}(-1) - J_{\alpha+h,\beta}(r)^{-1}} \right|. \end{aligned} \tag{B.5}$$

All square roots in the above expression are positive with non-negative real arguments. Moreover, the absolute values in Equation (B.5) make the logarithm arguments positive. For $1 \le C < -z$, we have the important formulas:

$$\mathsf{R}(r,h,C) = U_{\lambda_{T_{\mathrm{R}},C}}(\alpha), \qquad \mathsf{L}(r,h,C) = U_{\lambda_{T_{\mathrm{L}},C}}(-1). \tag{B.6}$$

We give a short analytic justification of the first identity in (B.6). The accompanying script `symbolic_endpoint_verifier.py` carries out the omitted partial-fraction, branch, differentiation, asymptotic, and endpoint calculations. It also verifies the second identity by the analogous computation.

Sketch of the first identity in (B.6). For the right compression, recall

$$T_{\mathrm{R}} = (\beta - h, \beta) = (2 - \beta, \beta)$$

and let

$$Q(w) := \sqrt{(w - \alpha)(w - (2 - \beta))},$$

where the branch is holomorphic on $\mathbb{C} \setminus [\alpha, 2 - \beta]$ and satisfies $Q(w)/w \to 1$ at infinity. Direct simplification of Equation (4.2) gives

$$S_{\mathrm{R}}(w) := S_{T_{\mathrm{R}},C}(w) = \frac{(w+C)Q(w)}{(w-z)(w-r)(w-\beta)}. \tag{B.7}$$

Using a Joukowski parametrization associated with $[\alpha, 2-\beta]$, the script constructs an explicit holomorphic function $\mathcal{P}$ on the upper half-plane. It verifies that all logarithms in the expression of $\mathcal{P}$ are taken on branches holomorphic there, that

$$\mathcal{P}'(w) = S_{\mathrm{R}}(w),$$

and that, as $w \to \infty$ within the upper half-plane,

$$\mathcal{P}(w) = \operatorname{Log} w + \log \frac{4}{2 - \beta - \alpha} + o(1). \tag{B.8}$$

To identify this primitive with the logarithmic potential, define

$$\mathcal{A}(w) := \int_{\mathbb{R}} \mathrm{Log}(w-s)\, d\lambda_{T_{\mathrm{R}},C}(s), \qquad \mathrm{Im}\, w > 0, \tag{B.9}$$

where Log is the principal logarithm. Since $w-s$ remains in the upper half-plane for every real s, this branch is available simultaneously on the whole support. Moreover,

$$\mathcal{A}'(w) = \int_{\mathbb{R}} \frac{d\lambda_{T_{\mathrm{R}},C}(s)}{w-s} = S_{\mathrm{R}}(w), \quad \mathrm{Im}\, w > 0,$$

by Equation (4.4), and

$$-\mathrm{Re}\, \mathcal{A}(w) = \int_{\mathbb{R}} \log \frac{1}{|w-s|}\, d\lambda_{T_{\mathrm{R}},C}(s). \tag{B.10}$$

Since $\lambda_{T_{\mathrm{R}},C}$ is a probability measure, $\mathcal{A}(w) = \mathrm{Log}\, w + o(1)$ at infinity. Hence $\mathcal{P} - \mathcal{A}$ is constant on the connected upper half-plane, and

$$\mathrm{Re}(\mathcal{P} - \mathcal{A}) = \log \frac{4}{2-\beta-\alpha}.$$

It follows that

$$U_{\lambda_{T_{\mathrm{R}},C}}(x) = -\log \frac{2-\beta-\alpha}{4} - \lim_{\varepsilon \downarrow 0} \mathrm{Re}\, \mathcal{P}(x+i\varepsilon) \tag{B.11}$$

whenever the boundary value is finite.

The script evaluates the boundary limit in Equation (B.11) at $x = \alpha$, keeping track of the real parts of all logarithms, and reduces the resulting expression exactly to the right-hand side of Equation (B.4). Therefore

$$U_{\lambda_{T_{\mathrm{R}},C}}(\alpha) = \mathsf{R}(r,h,C).$$

The degenerate case $T_{\mathrm{R}} = \varnothing$ follows by continuity. Indeed, each summand in the definitions of R and L extends continuously to $h = 0$. The same can be easily argued about $U_{\lambda_{T_{\mathrm{R}},C}}(\alpha)$ and $U_{\lambda_{T_{\mathrm{L}},C}}(-1)$ as well, using the same ideas in the proof of Lemma 5.5. □

The function $C \mapsto \mathsf{R}(r,h,C)$ is affine, and its unique zero is $C_0(r,h)$ as defined in Equation (6.3). We record the following important formulas at the extremal values $(r,h) = (r_*,0)$:

Lemma B.1. *At $(r,h) = (r_*,0)$,*

$$\mathsf{L}(r_*,0,C_0(r_*,0)) = 0, \qquad \frac{\partial}{\partial r}\mathsf{L}(r,0,C_0(r,0))\Big|_{r=r_*} = 0.$$

Proof. For the duration of the proof, abbreviate

$$q = q_*, \qquad p = p_*, \qquad u = u_*, \qquad v = v_*, \qquad r = r_*, \qquad z = z_*, \qquad D = r - z,$$

and put

$$C_* := C(q_*, u_*, v_*).$$

After setting $h = 0$ in the formulas (B.6) and using $r = X_q(u)$, $z = X_q(v)$, and $D = r - z$, one obtains the exact identities

$$R_*(C) := \mathsf{R}(r_*,0,C) = p\Lambda(q) + \frac{v(z+C)\Lambda(v) - u(r+C)\Lambda(u)}{D}, \tag{B.12}$$

and

$$L_*(C) - \frac{p-1}{p} R_*(C) = \frac{v(z+C)F_{\mathrm{L}}(q,v) - u(r+C)F_{\mathrm{R}}(q,u)}{pD}, \tag{B.13}$$

where $L_*(C) := \mathsf{L}(r_*,0,C)$. The script `symbolic_contact_verifier.py` checks these exact reductions, the formula for the zero $C(q,u,v)$, and the chain-rule reduction used in the derivative identity below.

We first prove the value identity. Expanding Equation (B.12) as an affine function of C, the coefficient of C is

$$\frac{v\Lambda(v) - u\Lambda(u)}{D} > 0.$$

Indeed, $D > 0$, and the map $y \mapsto y\Lambda(y)$ is strictly increasing on $(0,1)$, while $u < v$. Consequently, $R_*(C) = 0$ has a unique solution. Solving this equation gives

$$C = \frac{ru\Lambda(u) - zv\Lambda(v) - Dp\Lambda(q)}{v\Lambda(v) - u\Lambda(u)} = C(q,u,v) = C_*.$$

Since $C_0(r_*,0)$ is defined as the zero of $C \mapsto \mathsf{R}(r_*,0,C) = R_*(C)$, it follows that

$$C_0(r_*,0) = C_*. \tag{B.14}$$

At the contact triple,

$$F_{\mathrm{R}}(q_*,u_*) = 0, \qquad F_{\mathrm{L}}(q_*,v_*) = 0.$$

Hence the right-hand side of Equation (B.13) vanishes. Taking $C = C_*$, and using $R_*(C_*) = 0$, we obtain

$$\mathsf{L}(r_*,0,C_0(r_*,0)) = L_*(C_*) = 0.$$

We now prove the derivative identity. The point is to vary r while keeping q, $p = p(q)$, and D fixed. For r near r_*, set

$$z(r) := r - D, \qquad X_q(u(r)) = r, \qquad X_q(v(r)) = z(r),$$

with $u(r_*) = u_*$ and $v(r_*) = v_*$. Since $X_q'(u_*) \neq 0$ and $X_q'(v_*) \neq 0$, these functions are differentiable near r_*, and

$$u'(r) = \frac{1}{X_q'(u(r))}, \qquad v'(r) = \frac{1}{X_q'(v(r))}. \tag{B.15}$$

Define

$$\widehat{R}(r,C) := \mathsf{R}(r,0,C), \qquad \widehat{L}(r,C) := \mathsf{L}(r,0,C),$$

and

$$\widehat{H}(r,C) := \frac{v(r)(z(r)+C)F_{\mathrm{L}}(q,v(r)) - u(r)(r+C)F_{\mathrm{R}}(q,u(r))}{pD}.$$

The same exact algebraic identity as in Equation (B.13) holds along this local one-parameter family:

$$\widehat{L}(r,C) - \frac{p-1}{p}\widehat{R}(r,C) = \widehat{H}(r,C). \tag{B.16}$$

Let

$$c(r) := C_0(r,0).$$

Since $\widehat{R}(r,C)$ is affine in C with nonzero coefficient near r_*, the function $c(r)$ is differentiable there. By definition, $\widehat{R}(r,c(r)) = 0$, so Equation (B.16) gives

$$\widehat{L}(r,c(r)) = \widehat{H}(r,c(r)). \tag{B.17}$$

Differentiate Equation (B.17) at $r = r_*$. The chain rule gives

$$\frac{d}{dr}\widehat{H}(r,c(r)) = \partial_r\widehat{H}(r,c(r)) + \partial_C\widehat{H}(r,c(r))\,c'(r).$$

At $r = r_*$ and $C = C_*$,

$$\partial_C\widehat{H}(r_*,C_*) = \frac{v_*F_{\mathrm{L}}(q_*,v_*) - u_*F_{\mathrm{R}}(q_*,u_*)}{p_*D} = 0.$$

Thus the derivative of $c(r)$ makes no contribution. Moreover, when $\partial_r\widehat{H}$ is expanded, every derivative hitting the expression $v(r)(z(r)+C)$ or $u(r)(r+C)$ is multiplied by either F_L or F_R, therefore also vanishes at $r = r_*$. The only surviving terms come from differentiating F_L and F_R. Using Equation (B.15), we obtain

$$\begin{aligned}\frac{d}{dr}\widehat{L}(r, c(r))\Big|_{r=r_*} &= \frac{1}{p_* D}\left(\frac{v_*(z_* + C_*)\partial_v F_{\mathrm{L}}(q_*, v_*)}{X'_{q_*}(v_*)} - \frac{u_*(r_* + C_*)\partial_u F_{\mathrm{R}}(q_*, u_*)}{X'_{q_*}(u_*)}\right)\\ &= \frac{K(q_*, u_*, v_*)}{p_* D} = 0.\end{aligned}$$

The last equality is precisely the third equation in the contact system. Since $\widehat{L}(r, c(r)) = \mathsf{L}(r, 0, C_0(r, 0))$, this proves the second assertion. □

Appendix C. Verification scripts and running instructions

The verification bundle contains five certification Python scripts:

- `contact_existence.py`: the intermediate-value theorem certificate for Lemma A.1, and related reductions; it also certifies the rigorous enclosures for the constants p_*, α, r_*, z_*, and D that are needed in later scripts;
- `erdos1038_forcing_exact.py`: certifies the uniform three-atom potential inequalities, relevant numerical inequalities used in Lemma 3.2;
- `erdos1038_exact_endpoint_certificate.py`: certifies all inequalities in Lemma 6.1, including rigorous evaluation of Φ, using the parameter enclosures imported from `contact_existence.py`;
- `symbolic_endpoint_verifier.py`: provides exact symbolic checks supporting the derivation of the two formulas in Equation (B.6);
- `symbolic_contact_verifier.py`: exact symbolic checks used in Lemma B.1.

The three interval certificates use Python 3.13, `python-flint` 0.8.0, and 384-bit Arb precision [8]. The two symbolic scripts use SymPy 1.14 and exact rational function arithmetic. They may be run individually as follows:

```
python contact_existence.py
ERDOS1038_ARB_BITS=384 python erdos1038_forcing_exact.py
ERDOS1038_ARB_BITS=384 \
  python erdos1038_exact_endpoint_certificate.py --workers 1
python symbolic_endpoint_verifier.py
python symbolic_contact_verifier.py
```

Alternatively, `run_all_certificates.py` executes them in this dependency order, creates a local `logs/` directory, writes one log per script, and terminates at the first failure. Every pass/fail decision in an interval certificate is made from an outward-rounded Arb bound, while the symbolic scripts reduce the examined formulas to $0 = 0$. The file `README.md` records the certified statements and software versions.

References

[1] *pipeline-math: Prover-verifier pipelines for mathematical reasoning*, GitHub repository (2024), https://github.com/Pengbinghui/pipeline-math.

[2] T. F. Bloom, *Erdős Problem #1038*, The Erdős Problems website, https://www.erdosproblems.com/1038.

[3] Á. Elbert, Uber eine Vermutung von Erdős betreffs Polynome. I, Studia Sci. Math. Hungar. **1** (1966), 119–128.

[4] Á. Elbert, Uber eine Vermutung von Erdős betreffs Polynome. II, Studia Sci. Math. Hungar. **3** (1968), 299–324.

[5] P. Erdős, Extremal problems on polynomials, in Approximation Theory II, Academic Press, 1976, pp. 347–354.

[6] P. Erdős, F. Herzog, and G. Piranian, *Metric properties of polynomials*, J. Analyse Math. **6** (1958), 125–148.

[7] A. Fryntov and F. Nazarov, *New estimates for the length of the Erdős–Herzog–Piranian lemniscate*, in *Linear and Complex Analysis*, Amer. Math. Soc. Transl. Ser. 2, vol. 226, Amer. Math. Soc., Providence, RI, 2009, pp. 49–60.

[8] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, IEEE Trans. Comput. 66 (2017), 1281–1292.

[9] Ch. Pommerenke, *On metric properties of complex polynomials*, Michigan Math. J. **8** (1961), no. 2, 97–115.

[10] N. Sothanaphan, *Notes on the variational structure of minimizers for superlevel sets of logarithmic potentials*, March 19, 2026 version. https://github.com/HKUST-AARON/Erdos-Problem-1038/blob/8c767ac5eef7eb99461eb1cd15ddff502f6019a4/1038/erdos-problem-1038-inf-first-variation.pdf.

[11] T. Tao, *Superlevel sets of logarithmic potentials*, unpublished notes, December 27, 2025, https://terrytao.wordpress.com/wp-content/uploads/2025/12/erdos-1038-2.pdf.

[12] Open Science Framework files for Erdos 1308: https://osf.io/d6uq5/overview?view_only=e31128706d264b9ab0f08692d6702cec.

[13] T. Tao, *The maximal length of the Erdős–Herzog–Piranian lemniscate length in high degree*, arXiv:2512.12455, 2025, https://arxiv.org/abs/2512.12455.

[14] S. Wang, A Proposed Complete Solution to Erdős Problem 1038, https://github.com/ShouqiaoW/erdos/blob/main/1038/paper.pdf

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